STU-Mamba: A U-Mamba With Cross-Scale Alignment and Temporal Adaptive Context Gating for Satellite Image Time Series Segmentation

Jingyi Li, Nan Wang, Qingxi Wu, and Ran Tao

***Abstract*—Semantic segmentation of Satellite Image Time Series (SITS) is essential for monitoring Earth's surface dynamics, particularly in agricultural applications. However, existing methods often suffer from spatial misalignment due to downsampling and show limited adaptability to the spatiotemporal heterogeneity of crop phenology. To address this, we propose STU-Mamba, a novel architecture that integrates selective state space models (SSMs) with a U-shaped framework. For spatial modeling, we introduce a cross-scale aligned encoding mechanism that combines a Cross-Scale Alignment (CSA) module and Dual-Directional Mamba (DDM) blocks. This design preserves spatial details, mitigates geometric misalignment during encoding, and captures omnidirectional contextual dependencies. For temporal modeling, we present a Temporal Adaptive Context Gating (TACG) module. Within it, a Mamba block encodes long-range dependencies, and a context-aware gating network generates pixel-wise weights based on local spatial context to collapse the temporal dimension via adaptive aggregation. This enables the model to accurately characterize the phenological dynamics of the same crop type across different regions, effectively addressing the spatiotemporal heterogeneity inherent in agricultural landscapes. Finally, we construct a unified encoder–decoder framework entirely driven by Mamba operators, also embedding Mamba in the decoder for spatial refinement. Experiments on two public benchmarks, MTLCC and PASTIS, show that STU-Mamba achieves state-of-the-art performance, reaching 79.10% mean Intersection over Union (mIoU) on MTLCC and outperforming previous methods. Our approach delivers high accuracy and favorable scaling with respect to sequence length, making it suitable for large-scale SITS analysis.**

***Index Terms*—Cross-Scale Alignment (CSA), Mamba, Satellite Image Time Series (SITS), Semantic Segmentation, State Space Models (SSMs), Temporal Adaptive Context Gating (TACG).**

I. Introduction

ACCURATE and timely mapping of agricultural land cover is fundamental to global food security, sustainable resource management, and climate resilience planning [1]. Satellite Image Time Series (SITS), which captures the complete phenological evolution of surface targets, have become a pivotal data source for this task [2], [3]. Beyond crop monitoring, the ability to parse complex landscapes from SITS supports a wide range of geoscientific applications, from ecosystem service assessment to urban sprawl analysis, providing critical evidence for informed policy-making.

However, efficiently and accurately extracting discriminative features from SITS remains a core challenge. This requires a model capable of learning robust spatial representations to capture fine-grained contexts (e.g., field boundaries and textures), while simultaneously modeling long-range temporal dependencies to distinguish crops with similar spectra but distinct phenological rhythms.

A primary bottleneck in spatial modeling arises from the multi-scale feature construction process. Standard downsampling operations, such as strided convolutions, aggregate local patches into coarse-grained feature maps. This inevitably discards high-resolution details crucial for distinguishing adjacent fields and small crop parcels. More critically, downsampling disrupts the precise spatial correspondence between fine- and coarse-grained features, leading to geometric misalignment. Subsequent multi-scale fusion (e.g., via skip or lateral connections) without explicit positional realignment fails to recover these losses and may even amplify the misalignment. In SITS segmentation, where pixel-level accuracy governs agricultural parcel delineation, these issues directly manifest as blurred boundaries and the loss of small targets.

In the temporal domain, the task is further complicated by the pronounced spatiotemporal heterogeneity inherent to agricultural landscapes. Different crops follow distinct phenological calendars, and even the same crop can exhibit divergent growth trajectories across fields, due to local microclimate and farming practices. This implies that for the same crop growing in shaded areas, field ridges, or under varying soil conditions, the key phenological phases most informative for characterization may vary. Consequently, an effective model must perform spatially adaptive temporal aggregation, dynamically integrating key temporal information per pixel based on local context, rather than applying a uniform pooling operation across the entire image.

While deep learning has advanced the field, existing approaches face inherent limitations. Recurrent Neural Networks (RNNs) [4], [5], [6] explicitly model sequences but suffer from computational inefficiency [7] and gradient instability [8] over long series. Convolutional Neural Networks

Received 25 February 2026; revised 19 May 2026 and 15 June 2026; accepted 25 June 2026. Date of publication 29 June 2026; date of current version 20 July 2026. This work was supported in part by the Science for a Better Development of Inner Mongolia Program under Grant 2022EEDSKJXM003-2, in part by the National Natural Science Foundation of China under Grant 62471041, in part by Beijing Natural Science Foundation under Grant L247008, and in part by the National Key Research and Development Program of China under Grant 2022YFB3903404. *(Corresponding author: Nan Wang.)*

The authors are with the School of Information and Electronics, Beijing Institute of Technology, Beijing 100081, China (e-mail: jingyili@bit.edu.cn; wang_nan@bit.edu.cn; 3120230711@bit.edu.cn; rantao@bit.edu.cn).

Digital Object Identifier 10.1109/TGRS.2026.3708102

1558-0644 © 2026 IEEE. All rights reserved, including rights for text and data mining, and training of artificial intelligence and similar technologies. Personal use is permitted, but republication/redistribution requires IEEE permission.
See https://www.ieee.org/publications/rights/index.html for more information.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 08:46:36 UTC from IEEE Xplore. Restrictions apply.

(CNNs) [9], [10], [11] are constrained by local receptive fields, and their downsampling layers exacerbate spatial detail loss. The attention mechanism [12], [13], [14] enables dynamic focus on salient time steps but has quadratic complexity, limiting its scalability to long SITS. More fundamentally, their weights for temporal collapse are computed based on each pixel's own spatiotemporal features while ignoring local spatial context, which is inadequate for solving phenological heterogeneity.

Recently, State Space Models (SSMs), particularly Mamba [15], have emerged as a promising alternative for modeling long sequences with linear complexity [16], [17], [18]. Although initial adaptations in remote sensing validate this potential [19], [20], [21], directly applying Mamba to SITS segmentation does not resolve these core challenges. Spatially, Mamba's reliance on a fixed scanning order (e.g., rasterization) to serialize 2-D features captures limited directional dependencies and can disrupt intrinsic 2-D adjacency. Furthermore, sequence modeling is highly sensitive to the geometric fidelity of inputs. Spatially misaligned and detail-impoverished features from standard downsampling introduce ambiguous contexts along the scanning path. This may propagate and amplify local errors into global feature distortions, thereby disrupting the modeling of spatial relationships. Temporally, globally shared pooling strategies lack the spatial adaptability needed to model heterogeneous phenological dynamics across landscapes.

Compared with prior Mamba-based SITS methods (e.g., SITSMamba), our contribution is not merely replacing temporal encoders with Mamba, but a coordinated spatiotemporal design that jointly addresses cross-scale geometric alignment and context-aware temporal aggregation.

To address these limitations, we propose STU-Mamba, a unified framework designed for aligned cross-scale spatial modeling and spatially adaptive temporal aggregation in SITS. The main contributions are threefold.

1) We introduce a novel spatial modeling mechanism that combines a Cross-Scale Alignment (CSA) module with a Dual-Directional Mamba (DDM) block. The CSA module explicitly mitigates detail loss and spatial misalignment during downsampling through position-aligned fusion of fine- and coarse-grained features, providing geometrically faithful, detail-rich inputs. The DDM block then captures omnidirectional spatial context via row-wise and column-wise scanning followed by adaptive fusion, which is crucial for delineating irregular parcel boundaries.
2) We design a Temporal Adaptive Context Gating (TACG) module. Building upon Mamba-based temporal encoding, the TACG module aggregates local spatiotemporal context to generate pixel-wise adaptive weights for collapsing the temporal dimension. These weights are propagated across scales to gate skip connections, explicitly capturing heterogeneous phenological patterns across landscapes.
3) We construct STU-Mamba, a unified and scalable U-shaped architecture that integrates CSA-enhanced spatial encoder and the TACG-based temporal pathway. Centered on the Mamba operator with linear complexity in sequence length, it enables joint modeling of multi-scale spatiotemporal features from local details to global context.

Extensive experiments on the MTLCC [22] and PASTIS [12] benchmarks demonstrate that STU-Mamba achieves state-of-the-art performance, offering a potent and scalable solution for analyzing complex agricultural landscapes from SITS.

II. Related Works

A. Spatial Modeling: From Local Operators to Global Context

Constructing robust spatial representations is fundamental to remote sensing image interpretation. The field has evolved from reliance on handcrafted features to data-driven, deep learning paradigms. CNNs enable automatic feature learning through the inductive bias of local connectivity. Architectures like U-Net [23] achieve multi-scale fusion via encoder–decoder structures with skip connections, while Feature Pyramid Networks (FPNs) [24] explicitly build feature hierarchies through lateral connections. Other designs, such as HRNet [25], maintain high-resolution representations through parallel multi-resolution branches. In remote sensing, variants including FPN-ConvLSTM [26] and 3-D U-Net [27] have incorporated these mechanisms. However, CNNs are inherently constrained by fixed local receptive fields, and their reliance on stride-based downsampling sacrifices high-frequency spatial details. This results in blurred cross-scale correspondences and softened object boundaries, posing challenges for precise segmentation. Vision Transformer (ViT) [28] captures explicit long-range spatial dependencies through self-attention. Their adaptations for remote sensing, ranging from foundational segmentation models like SETR [29] and efficient hierarchical designs like Swin Transformer [30] to dedicated SITS models like TSViT [14], have advanced spatial context modeling. However, their quadratic complexity remains a fundamental bottleneck, limiting scalability on large-scale, high-resolution SITS datasets.

Recent advances in SSMs enable efficient global modeling: Vision Mamba [31] and VMamba [32] introduced bidirectional or cross-scanning to serialize 2-D images, and RSMamba [20] optimized it for remote sensing image classification via dynamic multi-path activation. Although these approaches overcome the limitation of fixed unidirectional scanning, the effectiveness of sequence modeling remains highly dependent on the spatial quality of input features. For SITS segmentation, inputs are often detail-impoverished and geometrically misaligned due to standard downsampling; under such conditions, even advanced scanning strategies struggle to reconstruct precise spatial relationships. Related efforts on multi-scale fusion also highlight the importance of high-quality cross-scale interaction for dense prediction, as exemplified by the U3M model [33].

To address this, we propose a synergistic mechanism combining a CSA module and a DDM block. The CSA module performs position-aligned feature fusion after downsampling,

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 08:46:36 UTC from IEEE Xplore. Restrictions apply.

actively mitigating detail loss and spatial misalignment to provide detail-rich inputs. Building upon this optimized representation, the DDM block captures omnidirectional spatial dependencies through row-wise and column-wise selective scanning followed by adaptive fusion, which is crucial for delineating irregular agricultural parcels.

B. Temporal Modeling: From Static Pooling to Weighted Aggregation

Temporal modeling in SITS aims to extract discriminative phenological dynamics from observational sequences. Early approaches, including vegetation index calculation [34], statistical algorithms like CCDC [35], and classical sequence models such as DTW [36], [37] and HMMs [38], relied on handcrafted features or strong data assumptions. While interpretable, these methods had limited capacity to model complex, high-dimensional temporal patterns in satellite imagery.

With the advent of deep learning, paradigms evolved from simplistic aggregation to learned temporal weighting. Initial strategies employed simple global pooling (e.g., mean or max) to collapse the temporal dimension [39], which risked obscuring brief yet critical phenological stages. RNNs, including Long Short-Term Memory (LSTM) and Gated Recurrent Unit (GRU) networks, introduced learned sequential modeling by explicitly capturing dependencies across time steps. Architectures such as ConvLSTM [40], U-BiConvLSTM [26], and U-ConvLSTM [27] integrated these modules within CNN-based networks for SITS analysis. However, their inherent sequential computation constrains training efficiency and parallelizability, while gradient instability degrades performance on long sequences.

Attention mechanisms revolutionized temporal aggregation by enabling dynamic weighting across time. Pioneering works such as U-TAE [12] and PSE-TAE [41], which embed lightweight temporal attention, and TSViT [14], which utilizes self-attention, achieve outstanding performance by focusing on key phenological stages. However, the attention weights of each pixel are typically derived solely from its individual temporal–spectral signature without explicitly integrating local spatial context. This limits adaptation to pixel-level spatiotemporal heterogeneity in agricultural landscapes, where optimal temporal signatures vary locally due to microclimate, soil, or farming practices. Notably, SITSMamba [21] introduces Mamba for temporal encoding in SITS, yet applies a globally shared temporal pooling after the Mamba block, falling short of context-aware, spatially adaptive aggregation. Beyond remote sensing, general time-series SSM research also highlights the necessity of task-specific interactive designs, as verified in cross-attention multi-scale SSMs for time-series prediction [42] and frequency-aware interactive Mamba frameworks [43].

To bridge this gap, we propose the TACG module. Following the Mamba-based temporal encoding, TACG employs a local spatial gate to compute pixel-wise weights conditioned on the surrounding spatiotemporal context. This design moves beyond globally shared or pixel-isolated aggregation, allowing the model to summarize phenological patterns in a manner that adapts to parcel-specific heterogeneity across agricultural landscapes.

C. Mamba in Remote Sensing

SSMs [17], [18], particularly Mamba [15], have emerged as a powerful paradigm for efficient long-sequence modeling, offering linear complexity and input-dependent selectivity. Their success in language modeling spurred rapid adoption in computer vision. Early efforts focused on adapting the selective scanning mechanism to 2-D image data, soon extending to dense prediction tasks such as medical image segmentation [44] and video understanding [45].

In remote sensing, Mamba's potential has been quickly validated across diverse applications. This includes ChangeMamba for change detection [19], MambaHSI for hyperspectral image classification [46], and SF-Mamba for small-object segmentation [47]. These studies highlight its ability to capture global context with computational efficiency suitable for high-resolution imagery. However, existing designs predominantly incorporate Mamba as a component within hybrid CNN- or Transformer-based frameworks, limiting its potential as a unified backbone for joint spatiotemporal modeling. Most works target single-temporal scenes or image-level classification. For the complex semantic segmentation task of SITS, which requires coordinating multi-scale spatial details and long-range temporal dependencies, a unified architecture centered on Mamba that deeply integrates spatiotemporal modeling is still lacking.

In contrast to existing Mamba-based SITS segmentation models such as SITSMamba, which primarily focus on temporal encoding followed by globally shared temporal aggregation, STU-Mamba explicitly addresses two orthogonal challenges that are critical for parcel-level segmentation: 1) cross-scale geometric misalignment introduced by downsampling (handled by CSA) and 2) spatially adaptive temporal aggregation under local phenological heterogeneity (handled by TACG). In addition, DDM complements the serialized scanning of Mamba by providing omnidirectional spatial context through row-wise and column-wise scans with adaptive fusion. Together, these modules form a coordinated spatiotemporal design rather than a direct architectural substitution of Mamba.

To realize this coordinated spatiotemporal design, our STU-Mamba constructs a U-shaped encoder–decoder architecture centered on the Mamba operator. Within this framework, the DDM blocks further refine spatial context via the aforementioned row and column scans with adaptive fusion, while the Mamba-based temporal encoding pathway is coupled with the TACG module to compute pixel-wise weights. This integrated implementation, combined with the CSA module for preserving spatial detail, ultimately establishes a scalable SSM-based solution tailored to the challenges of SITS segmentation.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 08:46:36 UTC from IEEE Xplore. Restrictions apply.

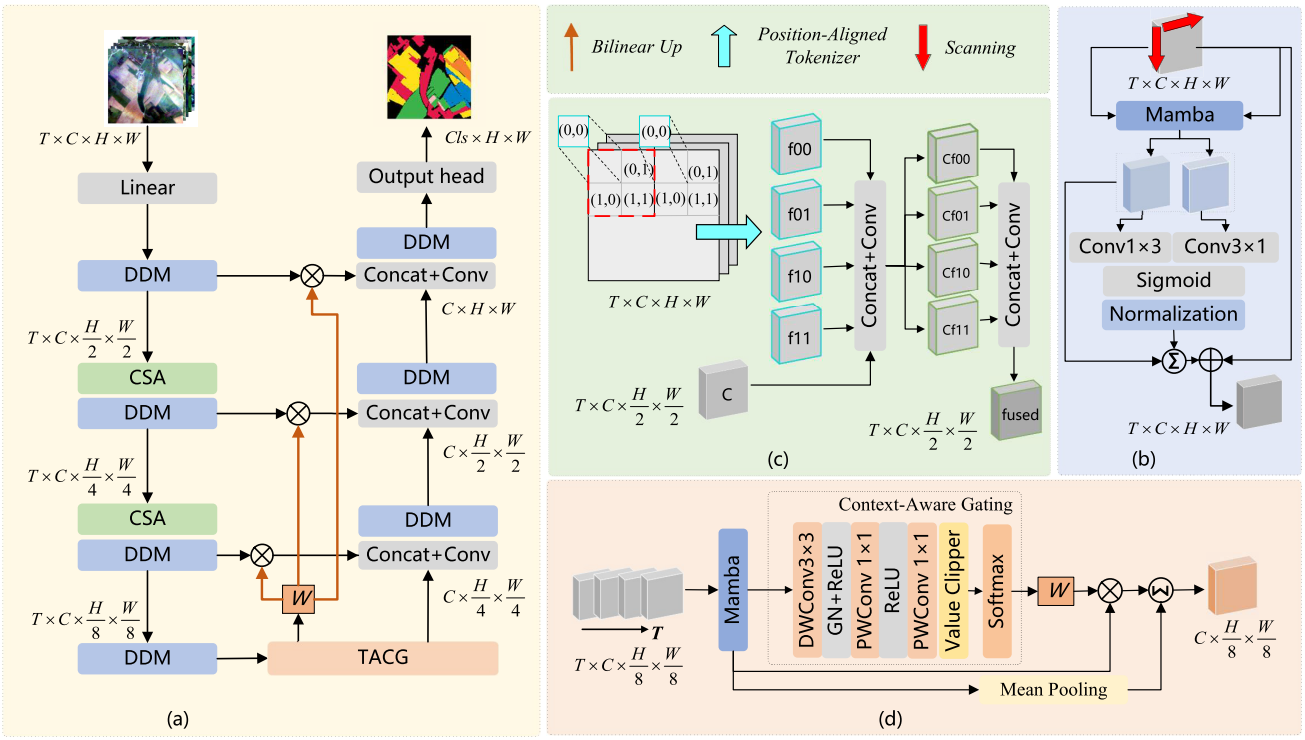


Fig. 1. Overview of the proposed **STU-Mamba** architecture. (a) U-shaped encoder–decoder framework. The spatial encoder constructs a multi-scale pyramid $\{\mathbf{F}_l\}$ using Cross-Scale Alignment (CSA) and Dual-Directional Mamba (DDM) blocks. The Temporal Adaptive Context Gating (TACG) module at the bottleneck generates pixel-wise adaptive weights $\mathbf{W}_{\text{temp}}$ to gate the multi-temporal skip connections $\mathbf{S}_l$ during decoding. (b) Dual-Directional Mamba (DDM) Block captures omnidirectional spatial context via row- and column-wise scanning with adaptive fusion. (c) Cross-Scale Alignment (CSA) Module performs position-aligned fusion between high-resolution fine tokens and low-resolution context, preserving details for scanning. (d) Temporal Adaptive Context Gating (TACG) Module produces pixel-wise temporal weights $\mathbf{W}_{\text{temp}}$ by aggregating local spatiotemporal context, enabling adaptive aggregation across heterogeneous fields.

III. Methodology

This section details the STU-Mamba architecture and its training strategy. We first present the overall design framework (see Section III-A), then provide an in-depth analysis of its three core components: the Cross-Scale Aligned Spatial Encoding Mechanism (see Section III-B), the TACG module (see Section III-C), the Decoder with Skip Connections (see Section III-D). Finally, we describe the composite loss function for model optimization (see Section III-E) and present a computational complexity analysis of the overall architecture (see Section III-F).

A. Overview Architecture

Semantic segmentation of SITS requires the simultaneous modeling of complex spatial details and long-range temporal dependencies. While existing U-Net architectures based on convolution-recurrence or convolution-attention mechanisms [12], [27], [48] have proven effective, they are constrained by local receptive fields or quadratic computational complexity. We propose STU-Mamba, a U-shaped encoder–decoder architecture centered on the Mamba operator, guided by two design principles: leverage Mamba's linear complexity and long-range dependency modeling for unified spatiotemporal processing; address the challenges of detail loss, spatial misalignment during downsampling, and the pronounced spatiotemporal heterogeneity in agricultural landscapes through targeted mechanisms.

The overall architecture is illustrated in Fig. 1. Given an SITS input tensor $\mathbf{X} \in \mathbb{R}^{T\times C\times H\times W}$, where T, C, H, and W denote the number of time steps, spectral channels, spatial height, and width, respectively, the model operates through three coordinated stages.

1) *Spatial Encoding:* A multi-scale encoder extracts a hierarchical feature pyramid $\{\mathbf{F}_l\}_{l=1}^{4}$ with channel dimensions [64, 64, 64, 128]. The encoder strategically deploys our Cross-Scale Aligned Spatial Encoding Mechanism across different stages to capture global dependencies while preserving fine-grained spatial details and geometric consistency.
2) *Temporal Encoding:* At the bottleneck, a Mamba block captures long-range dependencies within each pixel's time series. The TACG module then aggregates local context to generate a pixel-wise adaptive weight matrix $\mathbf{W}_{\text{temp}}$ for collapsing the temporal dimension of feature maps.
3) *Spatial Decoding:* A decoder progressively restores spatial resolution through transposed convolution and skip connections. Multi-temporal skip features are aggregated into single-temporal representations using the propagated weights $\mathbf{W}_{\text{temp}}$, which also gate cross-scale feature fusion.

Finally, a convolutional head maps the decoded features to the per-pixel semantic logits $\hat{\mathbf{Y}} \in \mathbb{R}^{N_{\text{cls}}\times H\times W}$.

B. Cross-Scale Aligned Spatial Encoding Mechanism

In constructing multi-scale feature pyramids, downsampling operations expand the receptive field but reduce spatial resolution, gradually losing fine details critical for segmentation. Classical skip connections only supplement details

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 08:46:36 UTC from IEEE Xplore. Restrictions apply.

during decoding and cannot prevent information loss along the encoding path. Moreover, for sequence-scanning models like Mamba, the richness and spatial accuracy of input features directly affect modeling performance. We therefore introduce a Cross-Scale Aligned Spatial Encoding Mechanism. After a strided convolution, the CSA module performs position-aligned, learnable fusion between the original fine-grained feature and the newly downsampled coarse features, explicitly injecting high-resolution information. The enhanced feature is then processed by the DDM block, which captures omnidirectional context through row- and column-wise bidirectional scanning followed by adaptive fusion. This synergistic design overcomes the coverage limitations of fixed unidirectional scans in irregular agricultural parcels.

1) CSA Module: The CSA module is deployed at critical transition points in the encoder to preserve fine-grained details through position-aligned cross-scale fusion. It takes two inputs: the fine-grained feature map $\mathbf{F}_{\text{fine}} \in \mathbb{R}^{C\times H\times W}$ from the previous layer and the coarsened feature map $\mathbf{F}_{\text{coarse}} \in \mathbb{R}^{C\times H/2\times W/2}$ obtained via a strided convolution, as illustrated in Fig. 1(c).

Instead of discarding $\mathbf{F}_{\text{fine}}$, the module first decomposes it into four non-overlapping sub-features corresponding to the 2×2 downsampling windows: $\{\mathbf{F}^{p}_{\text{fine}} \in \mathbb{R}^{C\times H/2\times W/2}\}$ for $p \in \{00, 01, 10, 11\}$. In practice, this partition is implemented by strided indexing over spatial positions: $\mathbf{F}^{00}_{\text{fine}}$ samples even rows and even columns, $\mathbf{F}^{01}_{\text{fine}}$ even rows and odd columns, $\mathbf{F}^{10}_{\text{fine}}$ odd rows and even columns, and $\mathbf{F}^{11}_{\text{fine}}$ odd rows and odd columns. This preserves exact positional correspondence with $\mathbf{F}_{\text{coarse}}$ at resolution $H/2\times W/2$. This 2×2 partition mirrors the stride-2 downsampling windows, enabling each coarse token to be fused with its exactly corresponding fine-grained context. For each index p, the coarse feature map $\mathbf{F}_{\text{coarse}}$ is concatenated with the corresponding fine sub-feature $\mathbf{F}^{p}_{\text{fine}}$ along the channel dimension and processed by a dedicated 1×1 convolution

$$\mathbf{F}^{p}_{\text{cross}} = \text{Conv}^{(p)}_{1\times 1}\left(\text{Cat}\left(\mathbf{F}_{\text{coarse}}, \mathbf{F}^{p}_{\text{fine}}\right)\right). \tag{1}$$

These 1×1 convolutions perform position-aligned channel mixing and compression (from $2C$ to C per branch), keeping the fusion lightweight while preserving strict geometric alignment. Each branch operates independently on its aligned pair, preventing cross-window mixing before the final projection.

Subsequently, the four cross-scale features are concatenated and integrated by a final projection layer to yield a single, detail-enhanced feature map

$$\mathbf{F}_{\text{fused}} = \text{Conv}^{(\text{proj})}_{1\times 1}\left(\text{Cat}\left(\mathbf{F}^{00}_{\text{cross}}, \mathbf{F}^{01}_{\text{cross}}, \mathbf{F}^{10}_{\text{cross}}, \mathbf{F}^{11}_{\text{cross}}\right)\right). \tag{2}$$

This final 1×1 convolution compresses the channel dimension from $4C$ back to C, completing the lightweight alignment.

The output $\mathbf{F}_{\text{fused}} \in \mathbb{R}^{C\times H/2\times W/2}$ is then fed into the subsequent DDM block. Notably, the CSA module performs alignment and fusion prior to serialization, so that only a single fused representation is fed into the DDM block. Although the module temporarily generates four position-aligned sub-features, this operation is conducted at the downsampled resolution of $H/2\times W/2$. After individual cross-scale fusion, the resulting feature maps are concatenated and immediately projected into a single output feature map, avoiding the duplication of high-resolution activations during the scanning process. By fusing independently downsampled coarse semantics with aligned fine-grained context before serialization, the CSA module ensures that the input sequences to the Mamba layers incorporate both semantic information at the current scale and detailed structural information from higher resolutions, thereby providing a richer and more spatially faithful foundation for subsequent modeling.

2) DDM Block: The DDM block, depicted in Fig. 1(b), is the core operator for selective state-space modeling on the spatially enhanced features. It addresses the limited contextual coverage of single-path scanning by modeling two orthogonal spatial directions and fusing them adaptively.

Given an input feature map $\mathbf{X} \in \mathbb{R}^{C\times H\times W}$, the block serializes it independently along the row and column directions. The row-wise scan flattens $\mathbf{X}$ in row-major order; the column-wise scan transposes $\mathbf{X}$ before flattening and transposes back after scanning. Each resulting 1-D sequence is processed by a Mamba operator, yielding two directional feature maps

$$\mathbf{Y}_{\text{row}} = \text{SiLU}\left(\text{GN}\left(\text{Mamba}\left(\text{Flatten}_{\text{row}}\left(\mathbf{X}\right)\right)\right)\right) \tag{3}$$

$$\mathbf{Y}_{\text{col}} = \text{SiLU}\left(\text{GN}\left(\text{Mamba}\left(\text{Flatten}_{\text{row}}\left(\mathbf{X}^{\top}\right)\right)\right)\right)^{\top} \tag{4}$$

where $\text{GN}(\cdot)$ and $\text{SiLU}(\cdot)$ denote the Group Normalization and the SiLU activation, respectively, with $\mathbf{Y}_{\text{row}}, \mathbf{Y}_{\text{col}} \in \mathbb{R}^{C\times H\times W}$. The row and column scanning order is chosen because the two orthogonal directions provide complementary long-range spatial context with moderate computational overhead, which is particularly suitable for delineating elongated and irregular agricultural parcels.

To fuse the two directional features, we introduce a lightweight gating network that generates spatially adaptive weights. We compute two gate maps using anisotropic convolutions followed by sigmoid activations:

$$\alpha_{\text{row}} = \sigma\left(\text{Conv}_{1\times 3}\left(\mathbf{Y}_{\text{row}}\right)\right), \quad \alpha_{\text{col}} = \sigma\left(\text{Conv}_{3\times 1}\left(\mathbf{Y}_{\text{col}}\right)\right). \tag{5}$$

They are normalized per pixel so that the two directional weights sum to one

$$\mathbf{G}_{\text{row}} = \frac{\alpha_{\text{row}}}{\alpha_{\text{row}} + \alpha_{\text{col}} + \epsilon}, \quad \mathbf{G}_{\text{col}} = 1 - \mathbf{G}_{\text{row}} \tag{6}$$

where ϵ is a small constant (e.g., 10^{-8}) for numerical stability.

The final output is obtained via adaptive fusion followed by a residual connection:

$$\mathbf{Y} = \mathbf{X} + \text{SiLU}\left(\text{GN}\left(\mathbf{G}_{\text{row}} \odot \mathbf{Y}_{\text{row}} + \mathbf{G}_{\text{col}} \odot \mathbf{Y}_{\text{col}}\right)\right) \tag{7}$$

where $\odot$ denotes the element-wise multiplication.

By fusing orthogonal scanning information based on local content, the DDM block dynamically balances contributions from different paths, maximizing long-range modeling effectiveness when operating on CSA-aligned inputs.

3) Multi-scale Encoder Implementation: Our spatial encoder constructs a four-level feature pyramid $\{\mathbf{F}_l\}_{l=1}^{4}$ via three strided convolutional downsampling operations, each using a 3×3 convolution with stride 2 and padding 1. For a multi-temporal input $\mathbf{X} \in \mathbb{R}^{T\times C\times H\times W}$, the batch and temporal dimensions are first flattened, enabling the shared encoder to process all $B\cdot T$ samples in parallel.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 08:46:36 UTC from IEEE Xplore. Restrictions apply.

Built around the DDM block as the core spatial modeling unit at each stage, the encoder employs a differentiated strategy using the CSA module for feature optimization.

1) *Stage 1:* Initial high-resolution features are first projected to 64 channels via a 1×1 convolution, then processed directly by a DDM block without CSA, as no downsampling has occurred.
2) *Stages 2 and 3:* After downsampling, the CSA module performs position-aligned fusion with higher-resolution features from the preceding stage to preserve spatial details, after which a DDM block processes the refined features.
3) *Stage 4:* At the lowest resolution, features are semantically rich and spatially condensed. Following downsampling, only a DDM block is applied for global context modeling, as explicit CSA is no longer required.

At stage l, the encoder transforms the input feature map $\mathbf{F}_{l-1,t}$ into $\mathbf{F}_{l,t}$ of size $C_l \times H_l \times W_l$ for each time step, where $H_l = H/2^{l-1}$ and $W_l = W/2^{l-1}$, and the channel dimension C_l follows the progression [64, 64, 64, 128] for $l = 1, 2, 3, 4$. These outputs are stacked temporally to form $\mathbf{F}_l \in \mathbb{R}^{T \times C_l \times H_l \times W_l}$, yielding a multi-scale feature pyramid $\{\mathbf{F}_1, \mathbf{F}_2, \mathbf{F}_3, \mathbf{F}_4\}$. Each level $\mathbf{F}_l$ encapsulates multi-temporal representations at the corresponding scale, ready for temporal fusion.

C. TACG Module

To obtain a single-temporal representation from multi-temporal encoder features, we apply a dedicated temporal module only at the bottleneck layer, following the efficient design of U-TAE [12]. This leverages rich semantics and a broad receptive field, and the generated temporal weights can be propagated across all scales via upsampling, avoiding the excessive cost of per-scale temporal modeling. However, U-TAE's attention mechanism suffers from quadratic complexity and, more critically, computes weights independently per pixel, relying solely on each pixel's own temporal profile while ignoring its local spatial context.

We therefore introduce the TACG module. A Mamba block first captures long-range temporal dependencies. A lightweight gating network then aggregates local spatiotemporal context to generate pixel-wise adaptive temporal weights. This allows the model to perceive local environmental cues and achieve truly context-aware temporal aggregation, which is essential for segmenting heterogeneous agricultural regions.

1) Temporal Modeling With Mamba: As shown in Fig. 1(d), the TACG module takes the lowest-resolution features $\mathbf{F}_L \in \mathbb{R}^{T \times C_L \times H_L \times W_L}$ from the spatial encoder as input. The channel dimension is first unified via a 1×1 convolution, and spatial dimensions are flattened to form independent temporal sequences. A Mamba block then captures global dependencies efficiently

$$\mathbf{Y}_{\text{seq}} = \text{Mamba}\left(\text{reshape}\left(\text{Conv}_{1\times1}\left(\mathbf{F}_L\right)\right)\right) \tag{8}$$

where $\mathbf{Y}_{\text{seq}} \in \mathbb{R}^{(H_L \cdot W_L) \times T \times d_{\text{model}}}$. Leveraging its selective state space mechanism and linear computational complexity, the Mamba block models the complete temporal evolution for each spatial location, outputting the enriched sequence $\mathbf{Y}_{\text{seq}}$.

2) Context-Aware Gating and Aggregation: To generate context-aware weights, a lightweight gating network $\mathcal{G}$ processes the Mamba output. The sequence $\mathbf{Y}_{\text{seq}}$ is first reshaped into a spatial-volume tensor $\mathbf{Y}_{\text{vol}} \in \mathbb{R}^{T \times d_{\text{model}} \times H_L \times W_L}$.

The network $\mathcal{G}(\cdot)$ consists of a depthwise separable convolution, group normalization, ReLU activation, and two 1×1 pointwise convolutions. The 3×3 depthwise convolution aggregates each pixel's local neighborhood features, allowing it to perceive critical local cues (e.g., field boundaries and shadow edges). The aggregated features are projected and compressed by subsequent pointwise layers, outputting spatial-adaptive raw scores for each spatiotemporal cell

$$\mathbf{A}_{\text{raw}} = \mathcal{G}\left(\mathbf{Y}_{\text{vol}}\right) \in \mathbb{R}^{T \times 1 \times H_L \times W_L}. \tag{9}$$

Finally, the pixel-wise temporal weight matrix is obtained by normalizing the clipped scores along the temporal dimension

$$\mathbf{W}_{\text{temp}} = \text{Softmax}\left(\text{clip}\left(\mathbf{A}_{\text{raw}},\ -10,\ 10\right)\right) \tag{10}$$

where $\text{clip}(\cdot,\ -10,\ 10)$ constrains values to $[-10, 10]$. This shifts the weighting paradigm from pixel-isolated to local context-aware.

3) Adaptive Aggregation and Multi-scale Propagation: Using weights $\mathbf{W}_{\text{temp}}$, we perform context-aware aggregation of the temporal sequences. For the i-th spatial position, the gated feature is computed as a weighted sum

$$\mathbf{z}_i = \sum_{t=1}^{T} w_{i,t} \cdot \mathbf{y}_{i,t} \tag{11}$$

where $\mathbf{y}_{i,t}$ denotes the t-th temporal feature vector at the i-th spatial position from $\mathbf{Y}_{\text{seq}}$.

To ensure robustness and mitigate potential over-reliance on the gating mechanism, we introduce a gated residual fusion strategy. The final representation for each pixel is a convex combination of the adaptively weighted feature and the global temporal average

$$\mathbf{z}_i^{\text{final}} = \lambda \cdot \mathbf{z}_i + (1 - \lambda) \cdot \frac{1}{T} \sum_{t=1}^{T} \mathbf{y}_{i,t} \tag{12}$$

where $\lambda \in [0, 1]$ is a learnable balancing parameter. This design ensures stability during training, especially for pixels with ambiguous local context.

The aggregated features are reshaped back to spatial dimensions and projected via a final 1×1 convolution to produce the bottleneck feature map $\mathbf{Z} \in \mathbb{R}^{C_{\text{out}} \times H_L \times W_L}$, which is passed to the decoder. Simultaneously, $\mathbf{W}_{\text{temp}}$ is retained and upsampled to guide the aggregation of multi-temporal skip connections across scales, as detailed in Section III-D.

In TACG, the temporal weight map $\mathbf{W}_{\text{temp}}$ is generated at the bottleneck resolution and bilinearly upsampled to match the spatial size of each skip level for (13). Note that bilinear interpolation is applied to the *temporal weights* rather than to the feature maps themselves. Consequently, the high-frequency spatial details of the skip features $\{\mathbf{F}_{l,t}\}$ remain intact, and the upsampled weights act as adaptive confidence scores that modulate the contribution of each time step without directly producing spatial details. Such mild smoothing introduced by upsampling does not significantly affect fine-grained

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 08:46:36 UTC from IEEE Xplore. Restrictions apply.

structures, because the weights remain spatially aligned after interpolation and are only used for pixel-wise temporal aggregation. Moreover, the subsequent decoder blocks further refine the fused representations, making the overall decoding robust to residual smoothing. This design allows TACG to provide context-aware, pixel-wise temporal emphasis across scales, capturing localized phenological dynamics that are typically overlooked by global shared pooling strategies.

D. Decoder With Skip Connections

The decoder consists of four symmetric stages, mirroring the encoder architecture, with channel dimensions progressing as [128, 64, 32, 32] from the bottleneck to the final output. Each stage performs transposed convolution upsampling, integrates gated skip connections, and refines spatial features. Each upsampling stage uses a transposed convolution with kernel size 4, stride 2, and padding 1 to double spatial resolution while reducing channel dimensions accordingly.

At level l (where $l = 4, 3, 2, 1$, with $l = 4$ being the bottleneck), the decoder takes two inputs: the upsampled feature $\mathbf{U}_{l+1} \in \mathbb{R}^{D_{l+1} \times H_l \times W_l}$ from the preceding deeper level, and the multi-temporal skip feature $\mathbf{F}_l \in \mathbb{R}^{T \times C_l \times H_l \times W_l}$ from the encoder, where D_l denotes the decoder channel dimension at level l. The process is initialized at the bottleneck with $\mathbf{U}_4 = \mathbf{F}_4 \in \mathbb{R}^{D_4 \times H_4 \times W_4}$.

The core operation is to aggregate the multi-temporal features into single-temporal representations for skip connections. Similar to U-TAE, we utilize the temporal weight matrix $\mathbf{W}_{\text{temp}}$ generated by the TACG module. This matrix is bilinearly interpolated to the spatial resolution of each decoder level l, yielding $\mathbf{W}_{\text{temp}}^{(l)} \in \mathbb{R}^{T \times 1 \times H_l \times W_l}$. The spatially adaptive skip feature $\mathbf{S}_l$ is obtained via weighted aggregation of $\mathbf{F}_{l,t}$ across all T time steps

$$\mathbf{S}_l = \sum_{t=1}^{T} \mathbf{W}_{\text{temp},t}^{(l)} \odot \mathbf{F}_{l,t} \tag{13}$$

where $\odot$ denotes the channel-wise broadcasting multiplication. Since $\mathbf{W}_{\text{temp}}$ is generated based on local spatiotemporal context, $\mathbf{S}_l$ retains spatial adaptivity even after interpolation.

Subsequently, $\mathbf{S}_l$ is concatenated with $\mathbf{U}_{l+1}$. The concatenated features are first aligned in channel dimension by a 1×1 convolution and then refined by a DDM block to enhance spatial coherence. The output of the current decoder level is, thus,

$$\mathbf{U}_l = \text{DDM}\left(\text{Conv}_{1\times1}\left(\text{Concat}\left(\mathbf{U}_{l+1}, \mathbf{S}_l\right)\right)\right). \tag{14}$$

Finally, the output of the last decoder stage, $\mathbf{U}_1 \in \mathbb{R}^{D_1 \times H \times W}$, is processed by an output head composed of two 3×3 convolutional layers to produce the per-pixel semantic prediction

$$\hat{\mathbf{Y}} = \text{ConvHead}\left(\mathbf{U}_1\right) \in \mathbb{R}^{N_{\text{cls}} \times H \times W} \tag{15}$$

where N_{cls} denotes the number of semantic classes.

E. Loss Function

To optimize the model for accurate per-pixel classification and geometrically coherent segmentation, we employ a composite loss function $\mathcal{L}$ combining the Cross-Entropy (CE) loss and the Lovász-Softmax loss

$$\mathcal{L} = \alpha \cdot \mathcal{L}_{\text{CE}} + \beta \cdot \mathcal{L}_{\text{LS}} \tag{16}$$

where α and β are the balancing coefficients.

The CE loss $\mathcal{L}_{\text{CE}}$ measures the pixel-wise classification error, providing strong gradients for discriminative feature learning

$$\mathcal{L}_{\text{CE}} = -\frac{1}{N} \sum_{n=1}^{N} \sum_{c=1}^{C} \mathbf{Y}_{n,c} \log\left(\hat{\mathbf{Y}}_{n,c}\right) \tag{17}$$

where N is the total number of pixels, C is the number of classes, $\mathbf{Y}$ is the ground truth, and $\hat{\mathbf{Y}}$ is the predicted probability.

The Lovász-Softmax loss $\mathcal{L}_{\text{LS}}$ directly optimizes the Jaccard index (IoU) for each class, effectively addressing class imbalance and improving boundary coherence

$$\mathcal{L}_{\text{LS}} = \frac{1}{C} \sum_{c=1}^{C} \bar{\Delta}_{J_c}\left(\mathbf{m}\left(c\right)\right) \tag{18}$$

where $\bar{\Delta}_{J_c}$ is the Lovász extension of the Jaccard loss for class c, and $\mathbf{m}(c)$ is the vector of errors for that class.

Empirically, we set $\alpha = 0.7$ and $\beta = 0.3$. This configuration ensures robust gradient propagation from the pixel-level CE loss while leveraging the Lovász loss's strength in promoting segmentation consistency and mitigating class imbalance in agricultural SITS data. The model parameters are updated by minimizing $\mathcal{L}$ over all training samples.

F. Computational Complexity Analysis

We analyze the overall computational complexity of the proposed STU-Mamba. Let the input be $\mathbf{X} \in \mathbb{R}^{T \times C \times H \times W}$. The encoder has $L = 4$ stages with resolutions (H_l, W_l) and channels C_l ($l = 1, \ldots, L$). Denote $N_l = H_l W_l$. Let d_{state} be the state dimension of the Mamba operator (a constant hyperparameter, e.g., 16) and d_{model} the feature dimension (e.g., 256).

1) Spatial Encoder With CSA and DDM: At stage l, the DDM block performs two directional selective scans (row/column) on spatial tokens. For each direction, the per-token cost includes: channel projections $\mathcal{O}(C_l^2)$, selective state-space scan $\mathcal{O}(C_l d_{\text{state}})$, and normalization/gating $\mathcal{O}(C_l)$. Processing T time steps independently and accounting for the bidirectional constant 2, the encoder complexity (excluding CSA) is $\mathcal{O}\left(T \sum_{l=1}^{L} N_l (C_l^2 + C_l d_{\text{state}} + C_l)\right)$.

For stages where CSA is enabled, the cross-scale 1×1 fusion introduces an additional $\mathcal{O}(T N_l C_l^2)$ term (constant number of 1×1 projections). Therefore, the encoder complexity can be written as

$$\mathcal{O}_{\text{enc}} = \mathcal{O}\left(T \sum_{l=1}^{L} N_l \left(C_l^2 + C_l d_{\text{state}} + C_l\right)\right) + \mathcal{O}\left(\sum_{l \in \mathcal{L}_{\text{CSA}}} T N_l C_l^2\right) \tag{19}$$

where $\mathcal{L}_{\text{CSA}}$ denotes the stages using CSA (typically $l = 2, 3$). In practice, the terms involving $C_l d_{\text{state}}$ and C_l are dominated

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 08:46:36 UTC from IEEE Xplore. Restrictions apply.

by the C_l^2 term because d_{state} is a small constant (e.g., 16) and C_l grows with the layer width.

2) TACG at the Bottleneck: At the bottleneck resolution $H_L \times W_L$, TACG processes N_L independent temporal sequences of length T with a Temporal Mamba. Its cost is dominated by channel projections: $\mathcal{O}(N_L T d_{\text{model}}^2)$. The context-aware gating network (depthwise separable convolutions) adds $\mathcal{O}(T N_L d_{\text{model}}^2)$, and the weighted aggregation in (11) costs $\mathcal{O}(N_L T d_{\text{model}})$. Thus, the bottleneck complexity is

$$\mathcal{O}\left(N_L T d_{\text{model}}^2\right). \tag{20}$$

3) Decoder With Skip Aggregation and DDM: For each decoder level $l \in \{1, \ldots, L-1\}$, the multi-temporal skip aggregation in (13) costs $\mathcal{O}(T \cdot N_l \cdot C_l)$. After aggregation, the decoder operates on single-temporal feature maps; thus, the DDM computation at level l costs $\mathcal{O}(N_l D_l^2)$, where D_l is the decoder channel width. Hence, the decoder complexity is

$$\mathcal{O}\left(\sum_{l=1}^{L-1} T N_l C_l\right) + \mathcal{O}\left(\sum_{l=1}^{L-1} N_l D_l^2\right). \tag{21}$$

4) Overall Complexity: The overall complexity is the sum of the encoder term in (19), the bottleneck TACG term in (20), and the decoder term in (21). For fixed architecture settings (fixed number of stages, fixed channel widths, and fixed hyperparameters such as d_{state} and d_{model}), all three components scale linearly with temporal length T and spatial token count $H \cdot W$. Hence, the overall complexity of STU-Mamba scales linearly with both T and $H \cdot W$ (up to fixed constant factors). The channel-mixing terms introduce C_l^2 and d_{model}^2 factors, but these are independent of the input dimensions and are constant for a given model configuration.

IV. Experiments

A. Datasets

We evaluate the performance of STU-Mamba and compare it with various baselines on two public SITS benchmarks: MTLCC [22] and PASTIS [12].

The MTLCC dataset [22] covers the region north of Munich, Germany, and contains 17 land-cover classes representative of Central European landscapes, including urban-rural transitions and agricultural zones. It provides 13 spectral bands by bilinearly interpolating four bands at 10 m, six bands at 20 m, and three bands at 60 m to a unified 10-m resolution. From the available 2016 imagery, we uniformly sample 30 time steps to cover a complete phenological cycle. Each input sample is a 48×48 pixel patch. We follow the official preset fold 0, splitting data into training, validation, and test sets in a 4:1:1 ratio while ensuring geographical independence and class balance.

The PASTIS dataset [12] encompasses four regions in southern France with diverse climates and cropping systems. It comprises 20 semantic categories, such as various crops, forest, urban, and water, including one "void" class for background or invalid regions. The dataset is constructed from Sentinel-2 observations between September 2018 and November 2019, containing 2433 samples. Each sample corresponds to a 128×128 pixel area at 10-m resolution, varying from 33 to 61 time steps and ten spectral bands per time series. To adapt to computational resources and improve training efficiency, we crop each sample into 64×64 sub-images. All experiments use only the first fold (fold-1) from the official five-fold split.

B. Implementation Details

All experiments are conducted on a single NVIDIA RTX 4090 GPU. Models are trained for 100 epochs with a batch size of 4, and an effective larger batch size is simulated using four-step gradient accumulation. To mitigate class imbalance, we adopt a hybrid loss function combining CE loss with a weight of 0.7 and Lovász loss with a weight of 0.3. For the variable-length sequences in PASTIS, padding and masking are applied to ignore padded values during loss computation and feature aggregation, preventing invalid information from interfering with model learning.

We use AdamW with weight decay set to 0.0. For PASTIS, we employ a learning rate schedule with warm-up and cosine annealing: an initial learning rate of 0.002, a linear warm-up over the first 12 epochs from 0 to the maximum, followed by cosine decay to $1e^{-5}$ by the end of training. For MTLCC, we use a fixed learning rate of 0.001, with all other training settings kept consistent with PASTIS.

We report standard evaluation metrics for semantic segmentation: Overall Accuracy (OA) and mean Intersection over Union (mIoU), defined as follows:

$$\text{OA} = \frac{1}{N} \sum_{i=1}^{N} \mathbb{I}(y_i = \hat{y}_i) \tag{22}$$

$$\text{IoU}_c = \frac{TP_c}{TP_c + FP_c + FN_c} \tag{23}$$

$$\text{mIoU} = \frac{1}{C} \sum_{c=1}^{C} \text{IoU}_c \tag{24}$$

where N is the total number of pixels, y_i and $\hat{y}_i$ are the ground truth and predicted labels of the i-th pixel, $\mathbb{I}(\cdot)$ denotes the indicator function, C is the number of valid classes excluding the void class, and TP_c, FP_c, and FN_c are the true positives, false positives, and false negatives for class c, respectively.

C. Comparison With State-of-the-Art Methods

To comprehensively assess performance on SITS semantic segmentation, we compare STU-Mamba with six representative state-of-the-art and classical methods. These baselines cover mainstream spatiotemporal feature extraction paradigms in current SITS processing, including CNN-based, RNN-based, Transformer-based, and emerging SSM-based methods. All comparative experiments are conducted on both MTLCC and PASTIS datasets under unified metrics and training settings for fair evaluation. Brief introductions to the compared methods are as follows.

1) *3-D U-Net [27]:* It employs 3-D convolutions to jointly model local spatial patterns and short-term temporal dynamics.
2) *UConvLSTM [27]:* It combines a U-Net encoder with ConvLSTM for spatiotemporal sequence modeling.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 08:46:36 UTC from IEEE Xplore. Restrictions apply.

TABLE I
QUANTITATIVE COMPARISON ON MTLCC AND PASTIS DATASETS. BEST RESULTS IN BOLD

Method	#Params	GFLOPs	Memory (MB)	IT (ms)	MTLCC		PASTIS	
					OA (%)	mIoU (%)	OA (%)	mIoU (%)
UConvLSTM	1.52M	12.06	169.7	13.26	93.47	72.87	82.78	61.32
FPN-ConvLSTM	1.15M	104.16	376.4	12.91	92.81	67.66	82.12	60.44
3D U-Net	1.55M	13.09	93.4	1.89	93.31	71.61	81.68	59.93
U-TAE	1.09M	11.64	168.1	8.65	94.16	76.73	83.32	63.60
TSViT	1.69M	11.59	78.1	5.53	92.55	71.20	81.45	62.49
SITSMamba	0.32M	27.81	578.9	5.93	94.58	74.70	82.69	63.72
STU-Mamba (Ours)	1.62M	15.01	373.4	26.29	**94.69**	**79.10**	**83.73**	**64.79**

Note: We report the number of parameters, GFLOPs, peak GPU memory, inference time (IT, ms per sample), overall accuracy (OA), and mean IoU (mIoU). GFLOPs, peak GPU memory and IT are measured on MTLCC with input size 48×48, $T = 30$ using a single NVIDIA RTX 4090 GPU. For Mamba-based models, GFLOPs are approximate lower bounds due to custom selective scan kernels.

3) *FPN-ConvLSTM [26]:* It integrates a FPN with bidirectional ConvLSTM for multi-scale spatiotemporal fusion.
4) *U-TAE [12]:* An attention-based SOTA model featuring a temporal attention encoder and a U-Net decoder.
5) *TSViT [14]:* A pure ViT that models SITS via self-attention across space and time.
6) *SITSMamba [21]:* A serial CNN-Mamba hybrid that applies convolutional layers followed by a Mamba block for spatiotemporal feature extraction.

1) Quantitative Results: Table I summarizes the quantitative comparisons. STU-Mamba achieves an excellent balance between parameter efficiency and accuracy. On the MTLCC dataset, which features moderate-length sequences, STU-Mamba attains 94.69% OA and 79.10% mIoU with 1.62M parameters. It outperforms the attention-based U-TAE (1.09M) by 2.37 mIoU percentage points and the Transformer-based TSViT (1.69M) by 7.90 mIoU points, highlighting the effectiveness of selective state-space modeling for capturing dependencies in such sequences.

On the more challenging PASTIS dataset, characterized by longer sequences and more diverse crop categories, STU-Mamba maintains leading performance with 83.73% OA and 64.79% mIoU. Benefiting from Mamba's linear complexity, it avoids the quadratic growth of self-attention with sequence length, offering favorable scaling characteristics for longer time series while achieving higher accuracy. Compared with U-TAE, STU-Mamba improves mIoU by 1.19 points, confirming that the TACG module better adapts to the spatiotemporal heterogeneity of complex agricultural landscapes. Relative to recurrent and convolutional baselines, it exceeds UConvLSTM (1.52M) by 3.47 mIoU points, 3-D U-Net (1.55M) by 4.86 points, and FPN-ConvLSTM by 4.35 points. Notably, it also outperforms another SSM-based model, SITSMamba, by 4.40 and 1.07 mIoU points on MTLCC and PASTIS, respectively, validating the advantage of our co-design spatial and temporal modeling over a serial CNN-Mamba pipeline.

In addition to accuracy, Table I reports model efficiency in terms of parameter count, computational complexity (GFLOPs), peak GPU memory, and single-sample inference time (IT) under a fixed input setting (48×48, $T = 30$). STU-Mamba achieves the best mIoU (79.10%) with 1.62M parameters, 15.01 GFLOPs, a moderate peak memory footprint of 373.4 MB, and an IT of 26.29 ms, demonstrating a favorable accuracy–efficiency trade-off. Although its GFLOPs and IT are slightly higher than those of U-TAE (11.64 G and 8.65 ms) and TSViT (11.59 G and 5.53 ms), this is mainly due to spatial granularity and boundary-oriented design. Specifically, TSViT uses patch-level tokenization [14] (patch size 4, reducing 48×48 pixels to $12 \times 12 = 144$ tokens), whereas STU-Mamba operates on the full pixel-level grid ($48 \times 48 = 2304$ spatial positions) to retain fine-grained spatial details critical for small and irregular agricultural parcels, which naturally increases the amount of computation and memory access per inference. In addition, the DDM block (bidirectional scanning and adaptive gated fusion) and the TACG module further contribute to accuracy improvements at a moderate increase in computational cost and latency. Meanwhile, STU-Mamba substantially reduces GFLOPs compared with heavier baselines such as SITSMamba (27.81 G) and FPN-ConvLSTM (104.16 G), while keeping a practical memory footprint for mainstream remote sensing GPUs. We note that for Mamba-based models, reported GFLOPs may be approximate lower bounds due to custom selective scan kernels; nevertheless, all methods are measured under the same protocol, and the overall efficiency trends are consistent.

2) Qualitative Results: Fig. 2 presents qualitative comparison results on the MTLCC dataset. We show four representative samples (a)–(d), including the ground-truth labels with parcel boundaries and crop categories, alongside the predictions from seven methods. Overall, STU-Mamba not only produces spatially consistent predictions over large-scale parcels but also accurately restores the outlines of scattered small parcels and intricate boundary details. In fragmented, irregular agricultural landscapes [e.g., sample (a)], STU-Mamba precisely reconstructs the shape and crisp boundaries of interlaced maize, meadow, hop, and winter barley parcels, outperforming all competing methods. For extensively contiguous hop parcels in sample (b), where other models exhibit obvious cross-class leakage and parcel under-coverage, STU-Mamba produces predictions free of internal misclassifications and maintains high homogeneity with the ground truth. Notably, only STU-Mamba accurately identifies the narrow winter triticale parcel surrounded by winter wheat

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 08:46:36 UTC from IEEE Xplore. Restrictions apply.

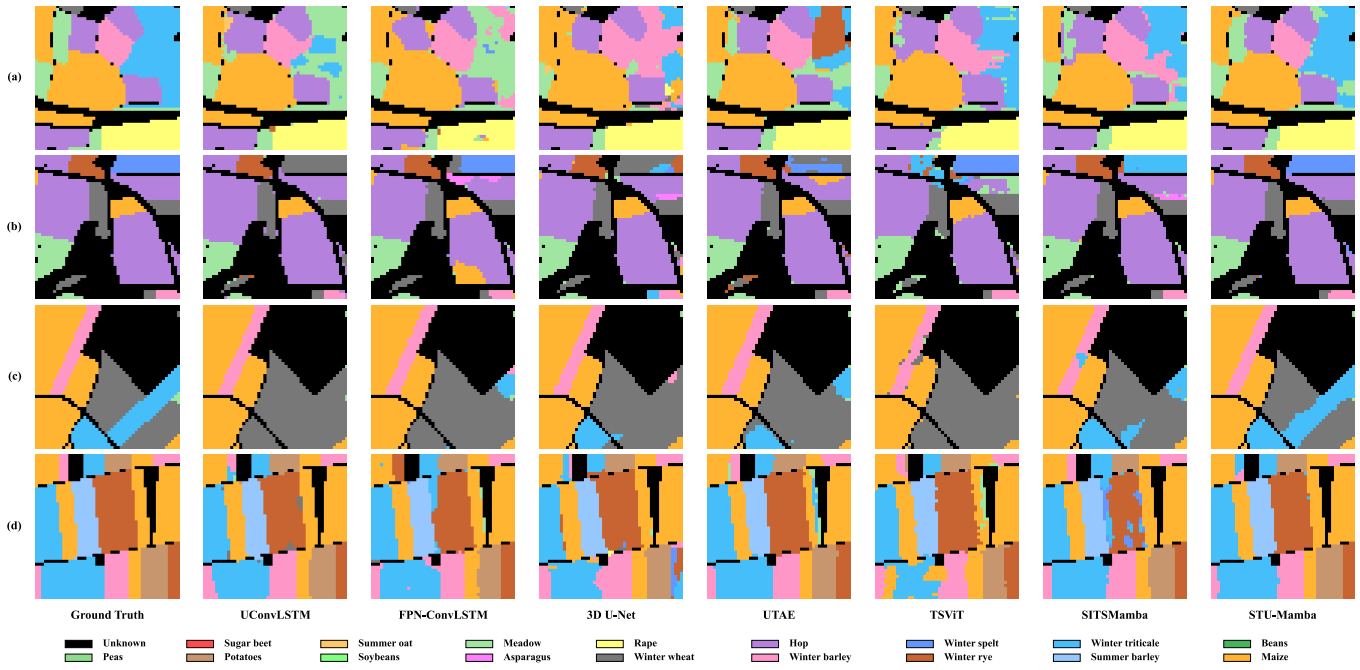


Fig. 2. Qualitative comparison on the MTLCC dataset. Four representative samples (a)–(d) are selected to illustrate the qualitative performance across diverse agricultural landscapes.

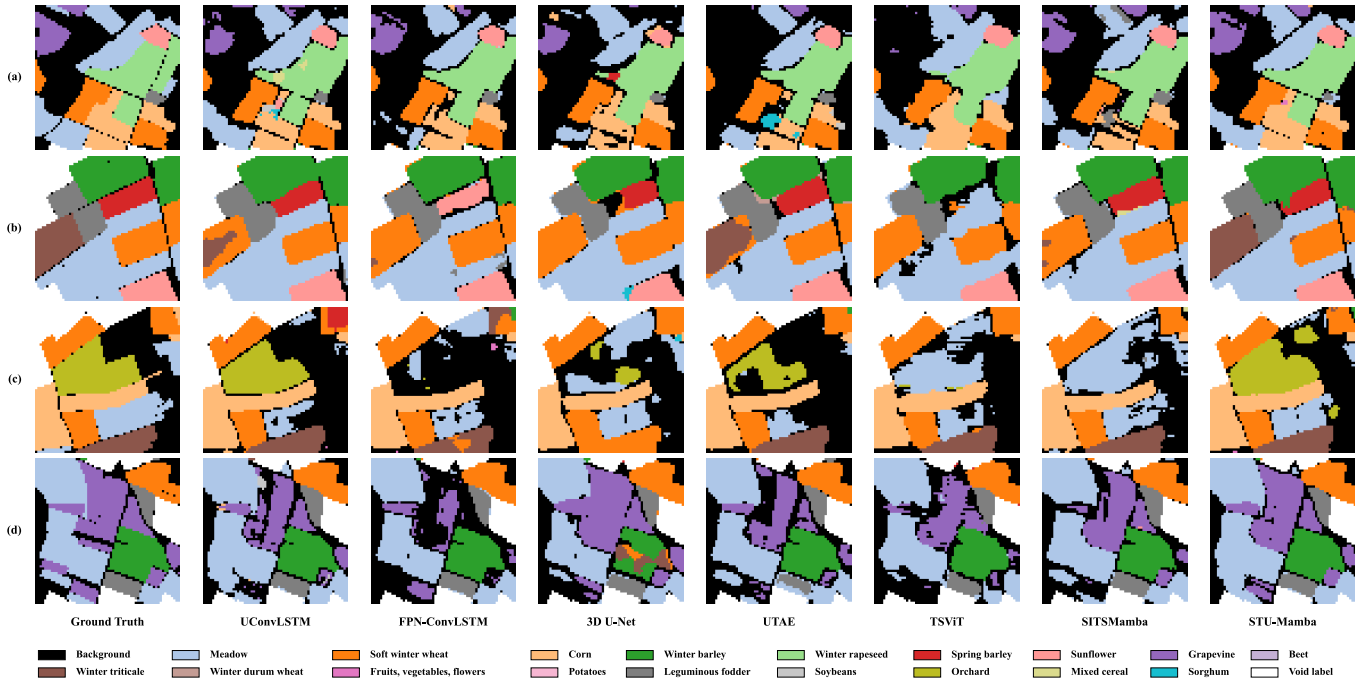


Fig. 3. Qualitative comparison on the PASTIS dataset. Four representative samples (a)–(d) are selected to demonstrate the qualitative performance across diverse agricultural landscapes.

in sample (c), effectively distinguishing crops with similar phenological patterns, whereas all other baselines misclassify it as the dominant surrounding crop. In mixed crop regions [e.g., sample (d)], STU-Mamba preserves the complete contours of parcels and fine-grained boundary information, which validates the synergistic effect of the CSA and DDM modules. Moreover, its accurate segmentation of the same crop classes (e.g., maize, hop, winter barley, and winter triticale) across disparate regions further demonstrates the capability of the TACG module in addressing inherent spatiotemporal heterogeneity.

Fig. 3 presents qualitative comparisons on the PASTIS dataset. STU-Mamba demonstrates excellent performance across diverse agricultural landscapes and can accurately

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 08:46:36 UTC from IEEE Xplore. Restrictions apply.

TABLE II
ABLATION STUDY OF STU-MAMBA ON MTLCC AND PASTIS DATASETS

Ablation Variant	MTLCC			PASTIS		
	OA (%)	mIoU (%)	Δ mIoU	OA (%)	mIoU (%)	Δ mIoU
STU-Mamba (Full)	94.69	**79.10**	—	83.73	**64.79**	—
w/o CSA Module	93.93	77.45	-1.65	83.07	63.86	-0.93
w/o Bidirectional Scanning (Row-only)	94.44	76.77	-2.33	83.60	64.24	-0.55
w/o TACG (Avg Pooling)	94.68	74.29	-4.81	83.04	60.45	-4.34
w/o Context Gating (Mamba + Avg)	94.26	77.41	-1.69	83.03	61.25	-3.54
Skip w/ Mean Aggregation	94.69	78.40	-0.70	82.93	61.85	-2.94

Note: "w/o" denotes removal of the component. TACG: Temporal Adaptive Context Gating module. CSA: Cross-Scale Alignment module. Δ mIoU values are computed relative to the full model.

distinguish complex crop types (e.g., winter rapeseed and corn). Constrained by the local receptive field of CNNs and the loss of details during layer-wise downsampling, UConvLSTM fails to preserve fine spatial edges of parcels. For instance, this model produces markedly fragmented predictions with salt-and-pepper noise for grapevine in sample (d). FPN-ConvLSTM exhibits both misclassification and under-segmentation, evident in the spring barley parcel of sample (b). 3-D U-Net segments the continuous winter barley parcels into multiple disjoint small parcels in sample (d), likely due to its reliance on local temporal convolutions, which struggle to capture global dependencies. Among the more competitive methods, TSViT yields blurry parcel boundaries lacking geometric sharpness. U-TAE exhibits inconsistent generalization ability, struggling to handle the spatiotemporal heterogeneity of the same crop across different parcels, showing significant degradation in identifying meadow in sample (c). SITSMamba, while preserving boundaries better than convolutional or recurrent baselines, still misses narrow, diagonally oriented parcels (e.g., the oblique grapevine patch in sample (d)), due to its unidirectional scanning scheme. By contrast, STU-Mamba retains the integrity of large homogeneous parcels while accurately recovering scattered small parcels with sharp boundaries, achieving an optimal balance between spatial precision and noise suppression. This highlights the superiority of the spatiotemporally unified Mamba architecture.

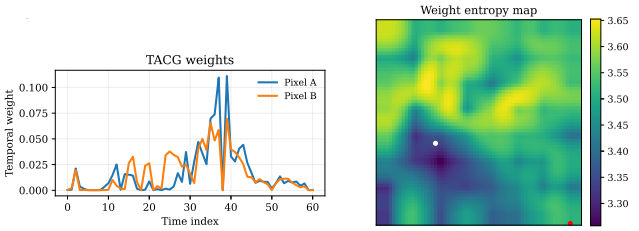


Fig. 4. Class 11 visualization example. (Left) TACG temporal weight curves. Interior Pixel A forms a sharp, dominant peak, while boundary Pixel B presents a flat, uniform distribution. (Right) Entropy map: low entropy (dark tones) corresponds to parcel interiors, and high entropy (bright tones) marks boundary zones. White dot denotes Pixel A, and red dot denotes Pixel B.

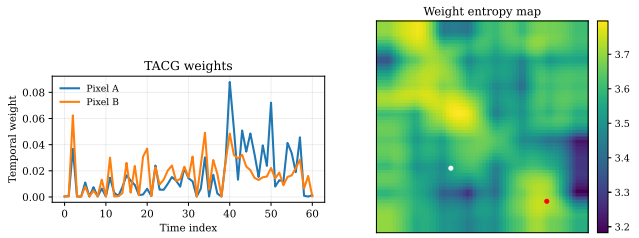


Fig. 5. Class 8 visualization example. (Left) TACG temporal weight curves. Interior Pixel A exhibits a sharp, high peak, whereas boundary Pixel B shows a flatter, evenly spread distribution. (Right) Entropy map with low entropy (dark) inside parcels and high entropy (bright) at boundaries. White dot denotes Pixel A, and red dot denotes Pixel B.

D. Ablation Study

We quantify the contribution of each core component by sequentially removing or replacing target modules on both the MTLCC and PASTIS datasets, using the full STU-Mamba model as the baseline. As summarized in Table II, model performance is evaluated by OA and mIoU, with the latter as the primary metric.

The results identify the TACG module as the most critical design. Replacing the entire temporal encoding pathway with simple average pooling leads to the most severe performance degradation, with mIoU dropping by 4.81 percentage points on MTLCC and 4.34 points on PASTIS. This confirms that the model can identify the importance of different time steps and focus on key phenological stages (e.g., crop heading or maturation), highlighting the necessity of dedicated temporal modeling for SITS segmentation. When isolating the context-aware gating mechanism alone (Mamba + Avg pooling), the performance decline is more pronounced on PASTIS (−3.54 points) than on MTLCC (−1.69 points). This underscores that context-aware weighting is essential for aggregating temporal features across heterogeneous landscapes, particularly for longer sequences with complex phenological dynamics.

To better understand this behavior, we visualize the temporal gating weights and corresponding entropy maps on two representative crop-cover categories from the PASTIS dataset (see Fig. 4 for Class 11 and Fig. 5 for Class 8). For each category, we deliberately pick one interior pixel (Pixel A) and one adjacent boundary pixel (Pixel B) within the same crop type instead of picking pixels from different classes. The results consistently show that interior pixels receive highly concentrated temporal weights with distinct dominant peaks and low entropy values, while boundary pixels exhibit flatter, more dispersed weight distributions accompanied by higher entropy values. This phenomenon is explained by TACG's design: the

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 08:46:36 UTC from IEEE Xplore. Restrictions apply.

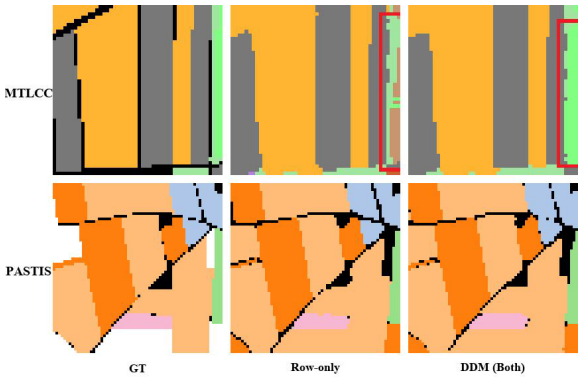


Fig. 6. Qualitative analysis of the DDM block under different parcel geometries. (Top) MTLCC: a selected example containing a narrow, elongated parcel. Removing the orthogonal scan (Row-only) breaks shape consistency and introduces local label confusion (the central part of the soybean parcel is misclassified as potatoes; red boxes), whereas the full DDM (Both) better preserves the elongated structure. (Bottom) PASTIS: a selected example with more compact and regular parcel geometry, where Row-only and DDM (Both) yield similar predictions.

depthwise separable convolution aggregates local spatial context. For interior pixels with unified and stable surroundings, TACG can efficiently screen out the most critical time-phase information; for boundary pixels under mixed spatial contexts, it perceives surrounding neighborhood features and generates more evenly distributed temporal attention weights. TACG achieves a larger performance improvement on PASTIS due to longer and more temporally heterogeneous sequences, whereas it yields a smaller gain on MTLCC with relatively shorter time series, as reported in Table II. These observations confirm that TACG aggregates local spatiotemporal context to accurately capture heterogeneous phenological patterns.

The spatial enhancements also provide consistent gains. Removing the CSA module reduces mIoU by 1.65 points on MTLCC and 0.93 points on PASTIS, validating its universal role in preserving spatial details during downsampling. The more pronounced effect on MTLCC is likely due to its lower input resolution (48×48 versus 64×64 for PASTIS), where the same downsampling operations cause relatively larger spatial misalignment, making the CSA module's alignment more critical.

Replacing the DDM block with unidirectional (Row-only) scanning causes a notably larger mIoU drop on MTLCC (-2.33 points) than on PASTIS (-0.55 points). We attribute this discrepancy mainly to spatial geometry. Specifically, row/column directional modeling provides complementary long-range context that is important for preserving shape continuity when parcels are narrow, elongated, or have dense boundaries. Fig. 6 provides qualitative evidence. On the selected MTLCC example, removing the orthogonal scan leads to discontinuous predictions and local label confusion (the central part of the soybean parcel is misclassified as potatoes; red boxes), while the full DDM (Both) better preserves the elongated parcel structure. In contrast, on the selected PASTIS example, where parcels appear more compact and regular, Row-only and DDM (Both) produce visually similar predictions, consistent with the much smaller average drop. These observations support that omnidirectional spatial context is particularly beneficial for complex parcel geometries, explaining the larger gain of DDM on MTLCC. The qualitative examples are provided for interpretation and complement the quantitative results.

Furthermore, replacing the adaptive weighted aggregation of multi-temporal skip connections with simple mean pooling results in a minor drop of 0.70 points on MTLCC, but a substantial drop of 2.94 points on PASTIS. This indicates that the learned temporal weights are crucial for effective multi-scale feature fusion, especially in longer time series.

In summary, the ablation study validates the effectiveness of each core component in STU-Mamba. The TACG module is central to handling spatiotemporal heterogeneity, the CSA and DDM modules enhance spatial modeling, and the adaptive gating and weighted skip-connection aggregation optimize temporal and multi-scale fusion. The performance impact of these components varies with datasets, such as sequence length, crop diversity, and spatial texture, yet their synergistic integration enables STU-Mamba to selectively capture the most critical spatiotemporal information, underpinning its robust performance.

V. Conclusion

In this article, we present STU-Mamba, a Mamba-based architecture that establishes an effective paradigm for SITS semantic segmentation through unified, position-aligned spatial modeling and spatially adaptive temporal aggregation. The proposed CSA module explicitly preserves fine-grained details and reduces cross-scale misalignment during downsampling, providing spatially faithful feature representations. Building on this, the DDM block captures omnidirectional spatial dependencies. Meanwhile, the TACG module addresses temporal heterogeneity by aggregating multi-temporal features via pixel-wise adaptive weights derived from local spatial context. Experiments on the MTLCC and PASTIS datasets demonstrate that STU-Mamba consistently outperforms existing methods. This validates its strengths in long-range dependency modeling, spatial detail preservation, and adaptive spatiotemporal aggregation. Although our method achieves promising segmentation accuracy, it still has limitations under extremely challenging conditions. When coping with scenes featuring highly blurred parcel boundaries and areas affected by severe shadow interference, the model may produce slight boundary deviations and occasional misclassifications. In the future, we will further strengthen the model's ability to distinguish fuzzy boundaries and improve its robustness against illumination variations. In follow-up research, we also plan to extend this Mamba-driven paradigm to other remote sensing tasks and explore its adaptability to more complex data scenarios, such as irregularly sampled or multimodal time series, to further enhance its practical applicability.

References

[1] C. Persello et al., "Deep learning and Earth observation to support the sustainable development goals: Current approaches, open challenges, and future opportunities," *IEEE Geosci. Remote Sens. Mag.*, vol. 10, no. 2, pp. 172–200, Jun. 2022.

[2] V. S. F. Garnot, L. Landrieu, S. Giordano, and N. Chehata, "Time-space tradeoff in deep learning models for crop classification on satellite multi-spectral image time series," in *Proc. IGARSS - IEEE Int. Geosci. Remote Sens. Symp.*, Jul. 2019, pp. 6247–6250.

[3] M. E. D. Chaves, M. C. A. Picoli, and I. D. Sanches, "Recent applications of Landsat 8/OLI and Sentinel-2/MSI for land use and land cover mapping: A systematic review," *Remote Sens.*, vol. 12, no. 18, p. 3062, Sep. 2020.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 08:46:36 UTC from IEEE Xplore. Restrictions apply.

[4] D. Ienco, R. Gaetano, C. Dupaquier, and P. Maurel, "Land cover classification via multitemporal spatial data by deep recurrent neural networks," *IEEE Geosci. Remote Sens. Lett.*, vol. 14, no. 10, pp. 1685–1689, Oct. 2017.

[5] M. Rußwurm and M. Körner, "Multi-temporal land cover classification with long short-term memory neural networks," *Int. Arch. Photogramm., Remote Sens. Spatial Inf. Sci.*, vol. 1, pp. 551–558, May 2017.

[6] M. Rußwurm and M. Körner, "Temporal vegetation modelling using long short-term memory networks for crop identification from medium-resolution multi-spectral satellite images," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. Workshops (CVPRW)*, Jul. 2017, pp. 1496–1504.

[7] A. Vaswani et al., "Attention is all you need," in *Proc. Neural Inf. Process. Syst.*, 2017, pp. 5998–6008.

[8] S. Hochreiter, "The vanishing gradient problem during learning recurrent neural nets and problem solutions," *Int. J. Uncertainty, Fuzziness Knowl.-Based Syst.*, vol. 6, no. 2, pp. 107–116, Apr. 1998.

[9] C. Pelletier, G. Webb, and F. Petitjean, "Temporal convolutional neural network for the classification of satellite image time series," *Remote Sens.*, vol. 11, no. 5, p. 523, Mar. 2019. [Online]. Available: https://www.mdpi.com/2072-4292/11/5/523

[10] S. Ji, C. Zhang, A. Xu, Y. Shi, and Y. Duan, "3D convolutional neural networks for crop classification with multi-temporal remote sensing images," *Remote Sens.*, vol. 10, no. 1, p. 75, Jan. 2018.

[11] M. Teimouri, M. Mokhtarzade, N. Baghdadi, and C. Heipke, "Fusion of time-series optical and SAR images using 3D convolutional neural networks for crop classification," *Geocarto Int.*, vol. 37, no. 27, pp. 15143–15160, Dec. 2022.

[12] V. S. Fare Garnot and L. Landrieu, "Panoptic segmentation of satellite image time series with convolutional temporal attention networks," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2021, pp. 4852–4861.

[13] P. Tang, P. Du, J. Xia, P. Zhang, and W. Zhang, "Channel attention-based temporal convolutional network for satellite image time series classification," *IEEE Geosci. Remote Sens. Lett.*, vol. 19, pp. 1–5, 2022.

[14] M. Tarasiou, E. Chavez, and S. Zafeiriou, "ViTs for SITS: Vision transformers for satellite image time series," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2023, pp. 10418–10428.

[15] A. Gu and T. Dao, "Mamba: Linear-time sequence modeling with selective state spaces," in *Proc. 1st Conf. Lang. Model.*, 2024. [Online]. Available: https://openreview.net/forum?id=tEYskw1VY2

[16] A. Gu et al., "Combining recurrent, convolutional, and continuous-time models with linear state-space layers," in *Proc. Neural Inf. Process. Syst.*, 2021.

[17] A. Gu, K. Goel, and C. Ré, "Efficiently modeling long sequences with structured state spaces," 2021, *arXiv:2111.00396*.

[18] J. T. H. Smith, A. Warrington, and S. W. Linderman, "Simplified state space layers for sequence modeling," 2022, *arXiv:2208.04933*.

[19] H. Chen, J. Song, C. Han, J. Xia, and N. Yokoya, "ChangeMamba: Remote sensing change detection with spatiotemporal state space model," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 4409720.

[20] K. Chen, B.-Y. Chen, C. Liu, W. Li, Z. Zou, and Z. X. Shi, "RsMamba: Remote sensing image classification with state space model," *IEEE Geosci. Remote Sens. Lett.*, vol. 21, pp. 1–5, 2024.

[21] X. Qin, X. Su, and L. Zhang, "SITSMamba for crop classification based on satellite image time series," in *Proc. IEEE Int. Geosci. Remote Sens. Symp. (IGARSS)*, Aug. 2025, pp. 7096–7099.

[22] M. Rußwurm and M. Körner, "Multi-temporal land cover classification with sequential recurrent encoders," *ISPRS Int. J. Geo-Inf.*, vol. 7, no. 4, 2018, Art. no. 129. [Online]. Available: https://www.mdpi.com/2220-9964/7/4/129

[23] O. Ronneberger, P. Fischer, and T. Brox, "U-Net: Convolutional networks for biomedical image segmentation," in *Proc. 18th Int. Conf. Med. Image Comput. Comput.-Assist. Intervent.*, vol. 9351. Cham, Switzerland: Springer, 2015, pp. 234–241.

[24] T. Y. Lin, P. Dollàr, R. Girshick, K. He, B. Hariharan, and S. Belongie, "Feature pyramid networks for object detection," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, Jul. 2017, pp. 2117–2125.

[25] J. Wang et al., "Deep high-resolution representation learning for visual recognition," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 43, no. 10, pp. 3349–3364, Oct. 2021.

[26] J. A. C. Martinez, L. E. C. La Rosa, R. Q. Feitosa, I. D. Sanches, and P. N. Happ, "Fully convolutional recurrent networks for multidate crop recognition from multitemporal image sequences," *ISPRS J. Photogramm. Remote Sens.*, vol. 171, pp. 188–201, Jan. 2021.

[27] R. M. Rustowicz, R. Cheong, L. Wang, S. Ermon, M. Burke, and D. Lobell, "Semantic segmentation of crop type in Africa: A novel dataset and analysis of deep learning methods," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR) Workshops*, Jun. 2019, pp. 75–82.

[28] A. Dosovitskiy et al., "An image is worth 16×16 words: Transformers for image recognition at scale," 2020, *arXiv:2010.11929*.

[29] S. Zheng et al., "Rethinking semantic segmentation from a sequence-to-sequence perspective with transformers," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2021, pp. 6877–6886.

[30] Z. Liu et al., "Swin transformer: Hierarchical vision transformer using shifted windows," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2021, pp. 10012–10022.

[31] L. Zhu, B. Liao, Q. Zhang, X. Wang, W. Liu, and X. Wang, "Vision Mamba: Efficient visual representation learning with bidirectional state space model," 2024, *arXiv:2401.09417*.

[32] Y. Liu et al., "VMamba: Visual state space model," in *Proc. Adv. Neural Inf. Process. Syst. 37*, 2024, pp. 103031–103063.

[33] B. Li, D. Zhang, Z. Zhao, J. Gao, and X. Li, "U3M: Unbiased multiscale modal fusion model for multimodal semantic segmentation," *Pattern Recognit.*, vol. 168, Dec. 2025, Art. no. 111801.

[34] S. Foerster, K. Kaden, M. Foerster, and S. Itzerott, "Crop type mapping using spectral–temporal profiles and phenological information," *Comput. Electron. Agricult.*, vol. 89, pp. 30–40, Nov. 2012. [Online]. Available: https://www.sciencedirect.com/science/article/pii/S0168169912002013

[35] Z. Zhu and C. E. Woodcock, "Continuous change detection and classification of land cover using all available Landsat data," *Remote Sens. Environ.*, vol. 144, pp. 152–171, Mar. 2014.

[36] V. Maus, G. Camara, R. Cartaxo, A. Sanchez, F. M. Ramos, and G. R. de Queiroz, "A time-weighted dynamic time warping method for land-use and land-cover mapping," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 9, no. 8, pp. 3729–3739, Aug. 2016.

[37] X. Xiao, L. Jiang, Y. Liu, and G. Ren, "Limited-samples-based crop classification using a time-weighted dynamic time warping method, Sentinel-1 imagery, and Google Earth engine," *Remote Sens.*, vol. 15, no. 4, p. 1112, Feb. 2023. [Online]. Available: https://www.mdpi.com/2072-4292/15/4/1112

[38] M. A. García, H. Moutahir, G. M. Casady, S. Bautista, and F. Rodríguez, "Using hidden Markov models for land surface phenology: An evaluation across a range of land cover types in Southeast Spain," *Remote Sens.*, vol. 11, no. 5, p. 507, Mar. 2019.

[39] N. Kussul, M. Lavreniuk, S. Skakun, and A. Shelestov, "Deep learning classification of land cover and crop types using remote sensing data," *IEEE Geosci. Remote Sens. Lett.*, vol. 14, no. 5, pp. 778–782, May 2017.

[40] X. Shi, Z. Chen, H. Wang, D.-Y. Yeung, W.-K. Wong, and W.-C. Woo, "Convolutional lstm network: A machine learning approach for precipitation nowcasting," in *Proc. Adv. Neural Inf. Process. Syst.*, vol. 28, 2015, pp. 802–810. [Online]. Available: https://proceedings.neurips.cc/paper_files/paper/2015/file/07563a3fe3bbe7e3ba84431ad9d055af-Paper.pdf

[41] V. S. F. Garnot, L. Landrieu, S. Giordano, and N. Chehata, "Satellite image time series classification with pixel-set encoders and temporal self-attention," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2020, pp. 12322–12331.

[42] D. Zhang, B. Li, F. Wang, Z. Zhao, J. Gao, and X. Li, "Cross-attention multi-scale state space model for remaining useful life prediction of aircraft engines," *Adv. Eng. Informat.*, vol. 69, Jan. 2026, Art. no. 103817.

[43] D. Zhang et al., "FAIM: Frequency-aware interactive Mamba for time series classification," 2025, *arXiv:2512.07858*.

[44] Z. Xing, T. Ye, Y. Yang, G. Liu, and L. Zhu, "SegMamba: Long-range sequential modeling Mamba for 3D medical image segmentation," 2024, *arXiv:2401.13560*.

[45] K. Li et al., "VideoMamba: State space model for efficient video understanding," in *Proc. Eur. Conf. Comput. Vis.* Cham, Switzerland: Springer, 2024, pp. 237–255.

[46] Y. Li, Y. Luo, L. Zhang, Z. Wang, and B. Du, "MambAHSI: Spatial–spectral Mamba for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5524216.

[47] M. Li, Z. Xing, H. Wang, H. Jiang, and Q. Xie, "SF-Mamba: A semantic-flow foreground-aware Mamba for semantic segmentation of remote sensing images," *IEEE MultimediaMag.*, vol. 32, no. 2, pp. 85–95, Apr. 2025.

[48] M. Papadomanolaki, M. Vakalopoulou, and K. Karantzalos, "A deep multitask learning framework coupling semantic segmentation and fully convolutional LSTM networks for urban change detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 9, pp. 7651–7668, Sep. 2021.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 08:46:36 UTC from IEEE Xplore. Restrictions apply.